chapter-07

Residue formula:

$$a_{-1} = \lim_{z \to a} \frac{1}{(k-1)!} \frac{d^{k-1}}{dz^{k-1}} \left\{ (z-a)^k f(z) \right\}$$

[k = order of poles
a = poles]

Exp: 01

Find the residues of $f(z) = \dfrac{z^2 - 2z}{(z+1)^2 (z^2+4)}$ at all poles in the finite plane.

$\Rightarrow$ **Poles:** $(z+1)^2 (z^2+4) = 0$ either $(z+1)^2 = 0$ or $(z^2+4) = 0$

$(z+1)^2 = 0$ $\qquad$ $z = \pm 2i$

$\therefore z = -1$

$z = -1$ is a $\underbrace{\text{double pole}}_{k=2}$ and

$z = \pm 2i$ are $\underbrace{\text{simple poles}}_{k=1}$

Residue at $(z = -1)$:

$$\text{Res}(f(z) = -1)$$

$$= \lim_{z \to -1} \frac{1}{(2-1)!} \cdot \frac{d^{2-1}}{dz^{2-1}} \left\{ (z+1)^2 \frac{z^2 - 2z}{(z+1)^2 (z^2+4)} \right\}$$

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$$= \lim_{z \to -1} d/dz \left(\frac{z^2 - 2z}{z^2+4} \right)$$

$$= \lim_{z \to -1} \frac{(z^2+4)(2z-2) - (z^2-2z)(2z)}{(z^2+4)^2}$$

$$= -14/25 \quad \text{(Ans)}$$

Residue at $(z = 2i)$:

$$\lim_{z \to 2i} \frac{1}{(1-1)!} \cdot \frac{d^{1-1}}{dz^{1-1}} \left\{ (z-2i) \frac{z^2 - 2z}{(z+1)^2 (z^2+4)} \right\}$$

$$= \lim_{z \to 2i} (z-2i) \frac{z^2 - 2z}{(z+1)^2 (z^2+4)}$$

$$= \lim_{z \to 2i} \cancel{(z-2i)} \frac{z^2 - 2z}{(z+1)^2 (z+2i) \cancel{(z-2i)}}$$

$$= \lim_{z \to 2i} \frac{z^2 - 2z}{(z+1)^2 (z+2i)}$$

$$= \frac{7+i}{25} \quad \text{(Ans)}$$

Residue at $(z = -2i)$:

$$\frac{7-i}{25} \quad \text{(Ans)}$$

Exp: 02

Evaluate $\oint \frac{z^n}{2z^2+5z+2}\,dz$ use the residue at the poles where $c: |z| = 1$

$\Rightarrow$

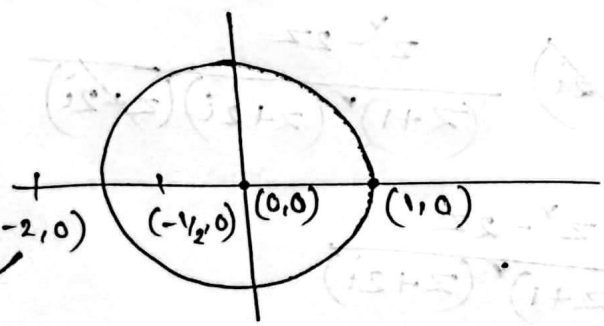
Poles: $2z^2+5z+2=0$

$\therefore z = -1/2, -2$

$\therefore z = -1/2, -2$ are simple poles.

$c: |z| = 1 \longrightarrow x^2+y^2 = 1$

As it outside, no need to find residue



Residue at $z = -1/2$:

$$\lim_{z \to -1/2} \frac{1}{(1-1)!} \left\{ \left(z+\tfrac{1}{2}\right)^2 \cdot \frac{z^n}{2z^2+5z+2} \right\}$$

$$= \lim_{z \to -1/2} \left((z+1/2) \frac{z^n}{(z+1/2)(z+2)} \right)$$

$$= \lim_{z \to -1/2} \frac{z^n}{z+2} = 1/6$$

(Ans)

Residue at $z=-2$:

$= 0$ [outside of c]

(d)

Using Residue Thm:

$$\oint f(z)\,dz = 2\pi i \sum \text{Res}$$

$$\therefore \int_c \frac{z^n}{2z^2+5z+2}\,dz = 2\pi i \left(\text{Res}(z=-1/2) + \text{Res}(z=-2)\right)$$

$$= 2\pi i \left(1/6 + 0\right)$$

$$= \pi i/3$$

(d)

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